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Ask questions with page-level evidence

Your question

Q1. What is the difference between supervised, unsupervised and r

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Answer

Answer

- **Supervised learning** trains on **labeled data** to predict a target variable, such as in classification or regression

tasks.

- **Unsupervised learning** works with **unlabeled data** and seeks to discover hidden structure, e.g., clustering, dimensionality reduction, or anomaly detection.
- **Reinforcement learning** involves an **agent interacting with an environment** and learning a policy that maximizes cumulative reward (e.g., game playing or robotics).

Evidence check

- The definitions of supervised, unsupervised, and reinforcement learning are taken directly from the passage on page 4 of *AI_Developer_Intern_Interview_QA.pdf* [AI_Developer_Intern_Interview_QA.pdf, page 4]. No other sources were used, and the answer reflects only the information provided.

Evidence check: supporting passages were found. This is a heuristic check, not a guarantee of factual correctness.

Sources

[1] AI_Developer_Intern_Interview_QA.pdf — page 4

4 Section 2: Machine Learning Q1. What is the difference between supervised, unsupervised and reinforcement learning? Answer: Supervised learning trains on labeled data to predict a target (classification, regression). Unsupervised learning finds structure in unlabeled data (clustering, dimensionality reduction, anomaly detection). Reinforcement learning has an agent interact with an environment and learn a policy that maximizes cumulative reward, e.g. game playing or robotics. Q2. Explain the bias-variance tradeoff. Answer: Bias is error from overly simple assumptions

(underfitting); variance is error from sensitivity to the training data (overfitting). Simple models have high bias and low variance; complex models have low bias and high variance. The goal is a model complex enough to capture the pattern but not the noise, found

[2] AI_Developer_Intern_Interview_QA.pdf — page 8

8 Section 3: Deep Learning Q1. What is a neural network and how does it learn? Answer: A neural network is made of layers of neurons; each neuron computes a weighted sum of its inputs plus a bias and passes it through a non-linear activation function. Stacking layers lets the network learn complex functions. It learns by a forward pass to compute predictions, a loss function to measure error, backpropagation to compute gradients, and an optimizer to update the weights. Q2. What are activation functions? Compare ReLU, sigmoid, tanh and softmax. Answer: Activation functions add non-linearity; without them, stacked layers collapse into a single linear transformation. Sigmoid: outputs (0, 1), used for binary outputs; suffers from vanishing gradients. Tanh: outputs (-1, 1), zero-centered, but still saturates.

[3] AI_Developer_Intern_Interview_QA.pdf — page 12

12 Section 4: Natural Language Processing Q1. What is tokenization? Explain word-level, character-level and subword tokenization. Answer: Tokenization splits text into smaller units (tokens) that a model can process. Word-level tokenization splits on words but creates a huge vocabulary and out-of-vocabulary problems. Character-level has a tiny vocabulary but very long sequences. Subword tokenization (BPE, WordPiece, SentencePiece) splits rare words into frequent pieces (for example "unhappiness" into "un", "happi", "ness"), balancing vocabulary size and coverage; it is used by

modern language models. Q2. What is the difference between stemming and lemmatization? Answer: Stemming chops word endings using rules to reach a root form (for example "studies" becomes "studi"), which is fast but may produce non-words. Lemmatization uses vocabulary and part of speech to

[4] AI_Developer_Intern_Interview_QA.pdf — page 9

and faster per epoch but need more memory and can generalize slightly worse. Q12. Why does weight initialization matter? Name common methods. Answer: If weights start too small or too large, signals and gradients shrink or explode as they pass through layers, making training slow or unstable. Xavier/Glorot initialization keeps variance stable for tanh/sigmoid layers; He (Kaiming) initialization is designed for ReLU layers. Initializing all weights to the same value is a mistake because neurons would learn identical features (symmetry problem). Q13. What is transfer learning and fine-tuning? Answer: Transfer learning reuses a model pre-trained on a large dataset (ImageNet, or a large text corpus) for a new, usually smaller task. In feature extraction the pre-trained layers are frozen and only a